

Fundamentals of Solid State Physics

Crystal Diffraction 晶体衍射

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Determination of Crystal Structures

- Crystallinity

- Single crystal? polycrystal? amorphous?



diamond?
glass?

- Lattice type

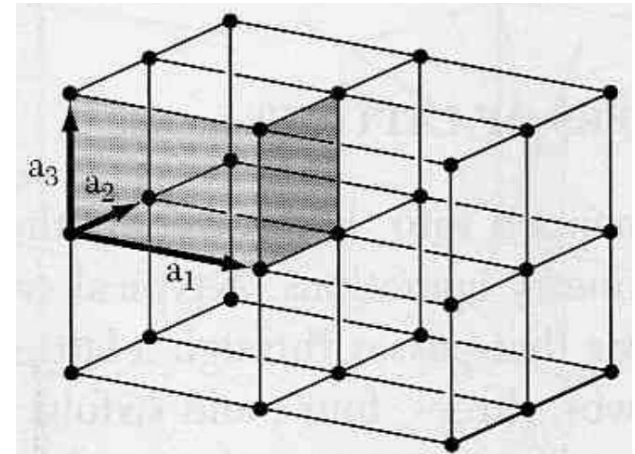
- BCC, FCC, ... ?

- Lattice parameter

- $a = ?$

- Other properties

- Defects? Crystal size? ...



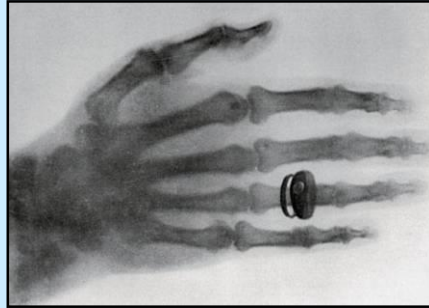
Determination of Crystal Structures

- **X-ray diffraction (XRD) X光晶体衍射**
 - **Bragg's law 布拉格定律**
 - **examples in cubic systems**
 - **XRD for polycrystals**

- **Other methods**
 - **electron diffraction**
 - **electron microscopy**
 - **neutron diffraction**
 - **...**

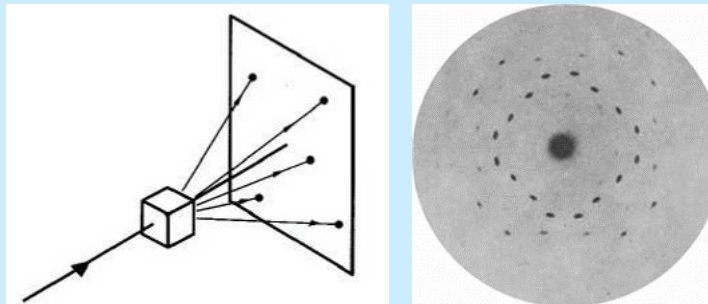
History

Discovery of X-ray



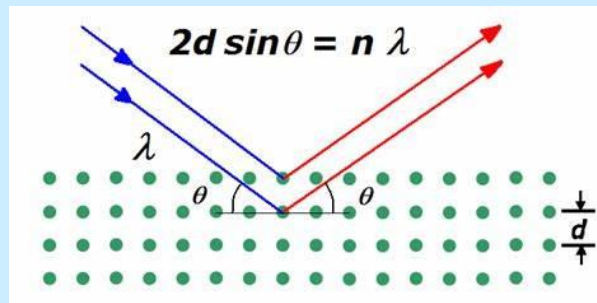
W. Rontgen (伦琴)
Nobel Prize in 1901

X-ray diffraction of crystals



M. von Laue (劳厄)
Nobel Prize in 1914

Bragg's law



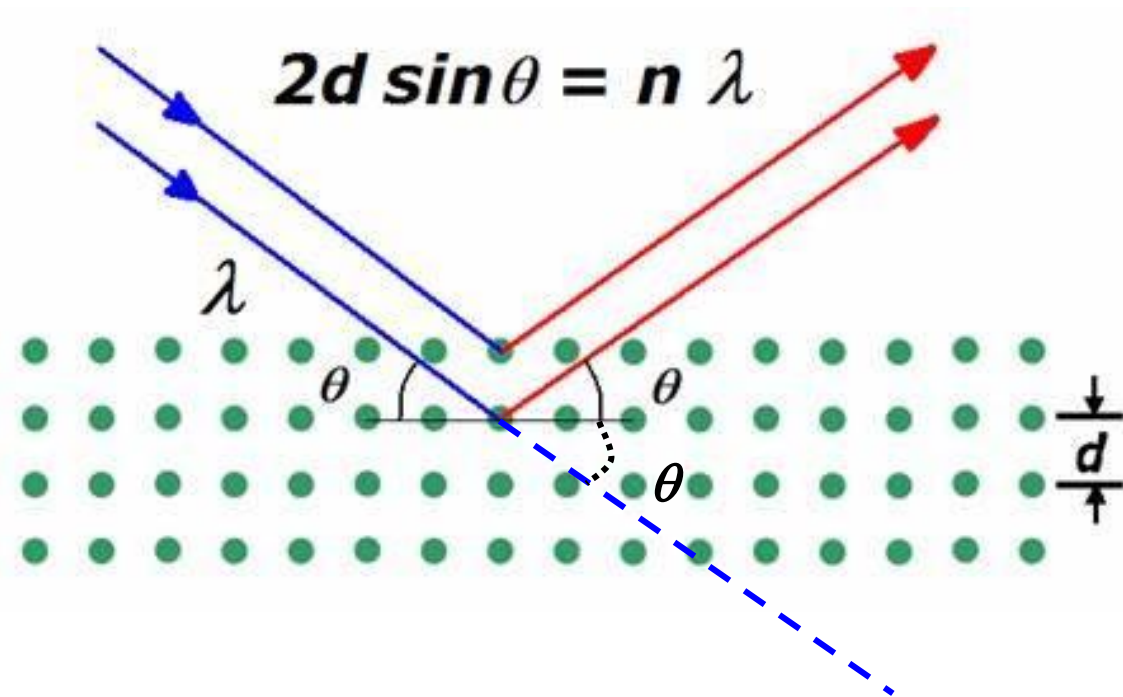
Bragg & Bragg
(布拉格父子)
Nobel Prize in 1915

Bragg's Law 布拉格定律

- X-ray diffraction (XRD) X光晶体衍射
 - constructive interference among each atomic layer

$$2d \sin \theta = n\lambda$$

d - interplanar spacing
 θ - Bragg angle
 2θ - Deflection angle
 λ - X-ray wavelength
 $n = 1, 2, 3, \dots$

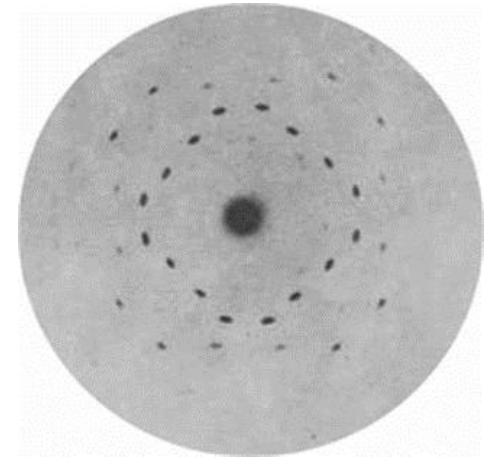
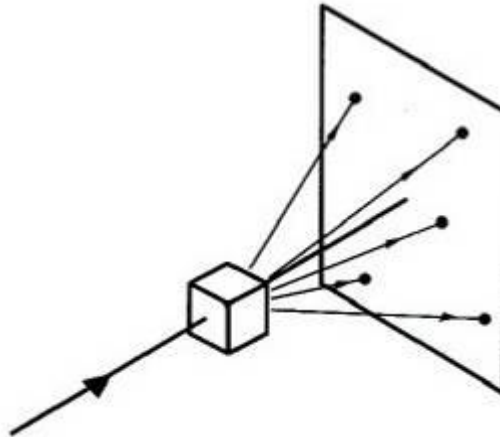


Typically, the first order diffraction ($n = 1$) has the strongest intensity.

Bragg's Law 布拉格定律

- X-ray diffraction (XRD) X光晶体衍射
 - constructive interference among each atomic layer

$$2d \sin \theta = n\lambda$$



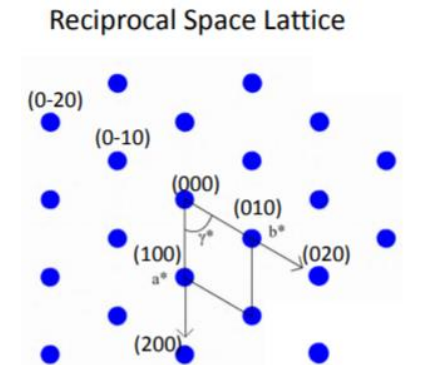
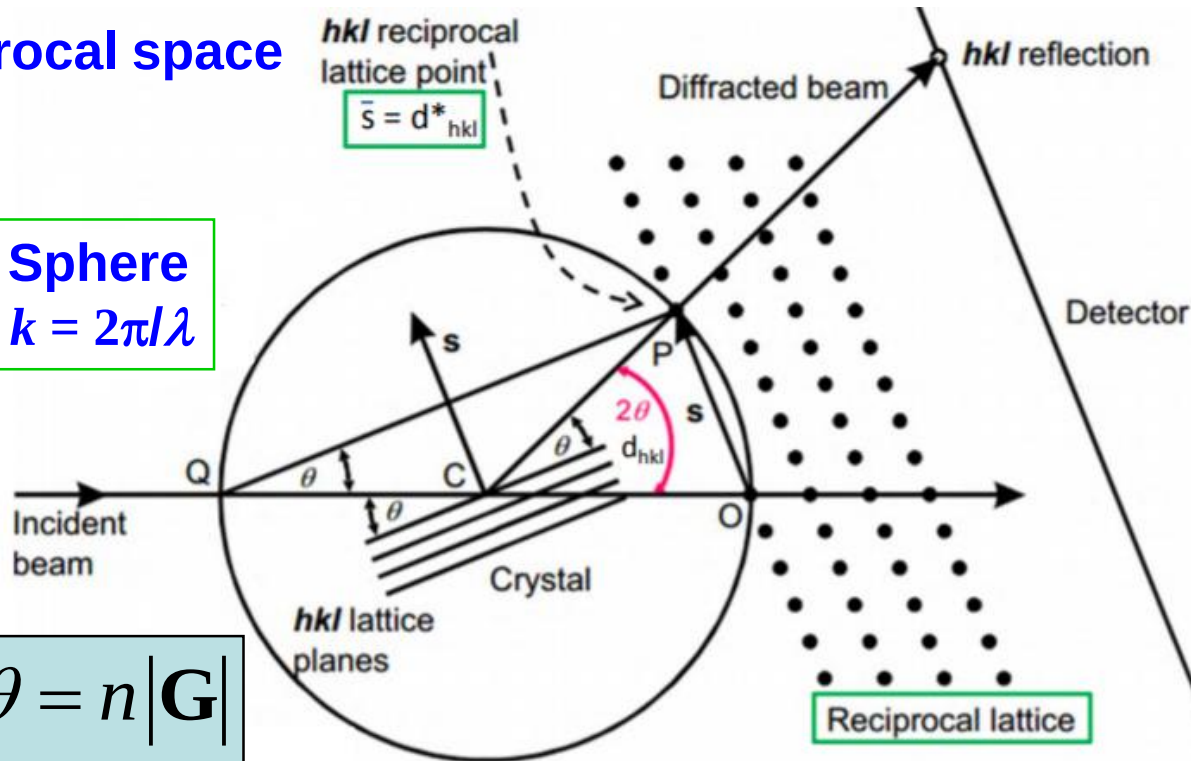
Diffraction Patterns of a single crystal

M. von Laue (劳厄)
Nobel Prize in 1914

von Laue Formulation

in reciprocal space

Ewald Sphere
radius $k = 2\pi/\lambda$



Ashcroft & Mermin
p.96–p.104

$$2k \sin \theta = n |\mathbf{G}|$$

$$2 \cdot \frac{2\pi}{\lambda} \cdot \sin \theta = n |\mathbf{G}| = n \frac{2\pi}{d_{(hkl)}}$$



$$2d_{(hkl)} \sin \theta = n\lambda$$

X-ray diffraction pattern is the reciprocal lattice of the crystal

Real Space vs. Reciprocal Space

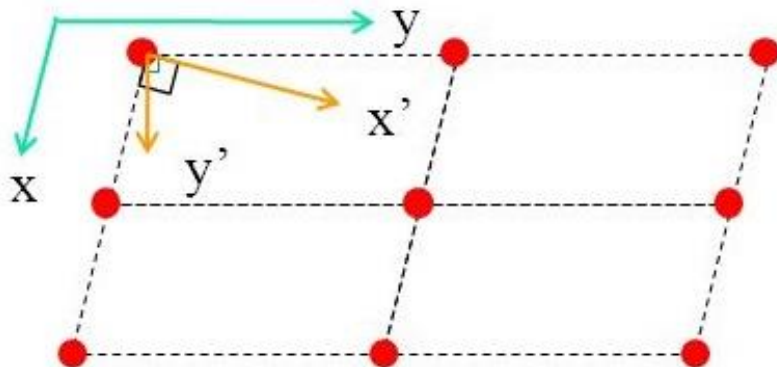
Crystal Structure



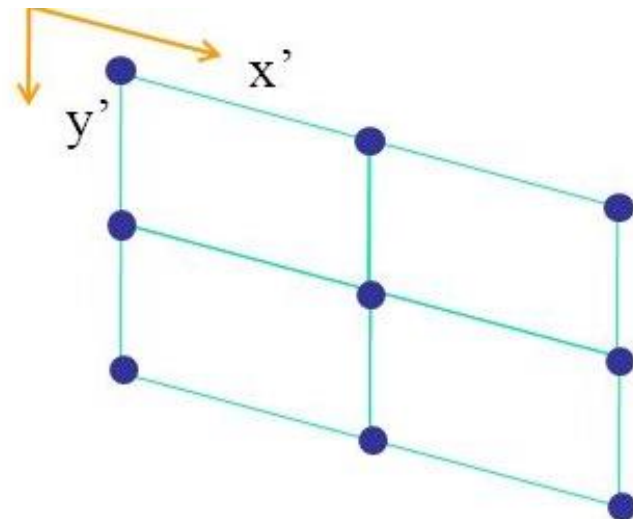
Diffraction Pattern

Direct Lattice

Reciprocal Lattice



Real Space

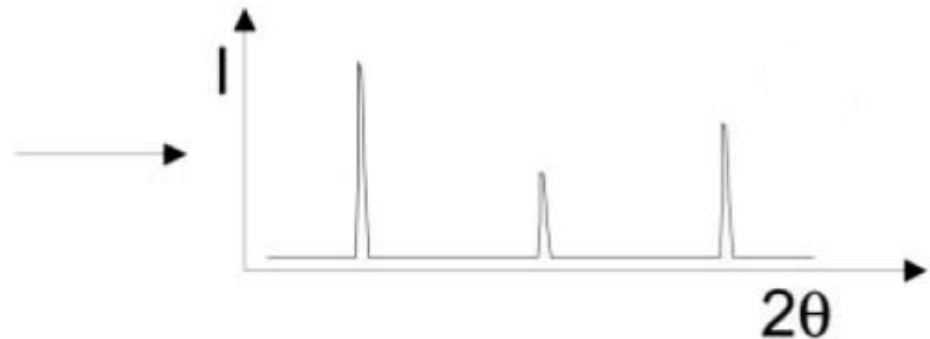
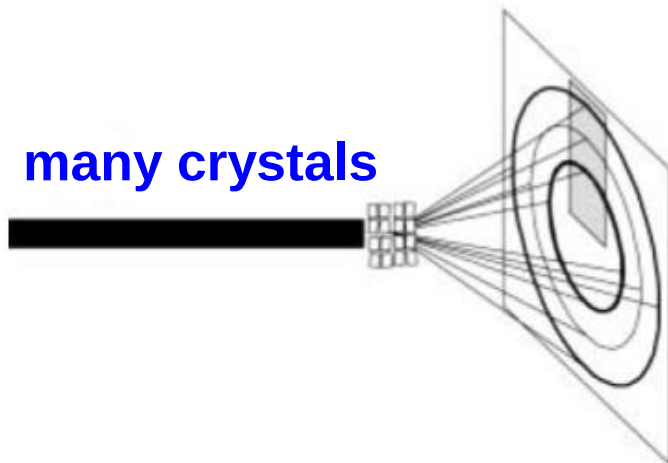
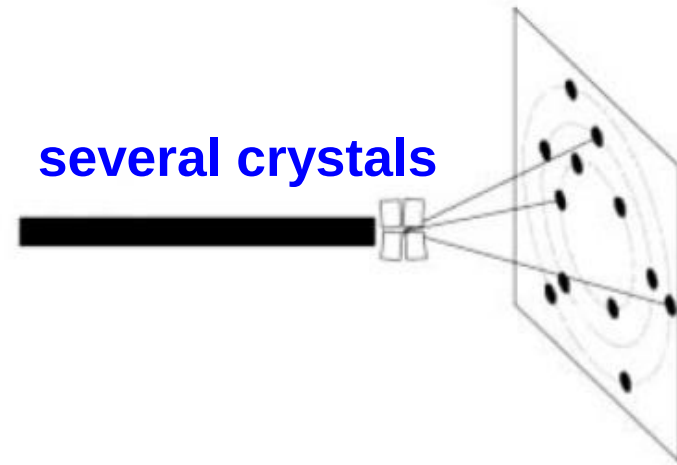
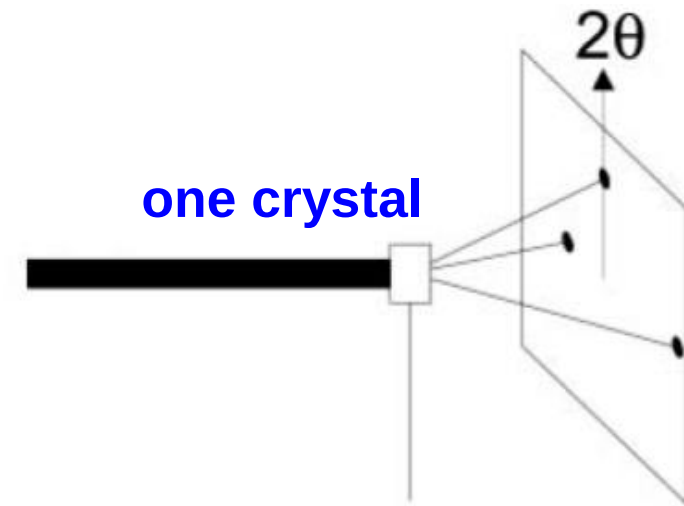


Reciprocal Space

X-ray diffraction pattern is the reciprocal lattice of the crystal

XRD of Crystals

$$2d \sin \theta = n\lambda$$



XRD of Polycrystals

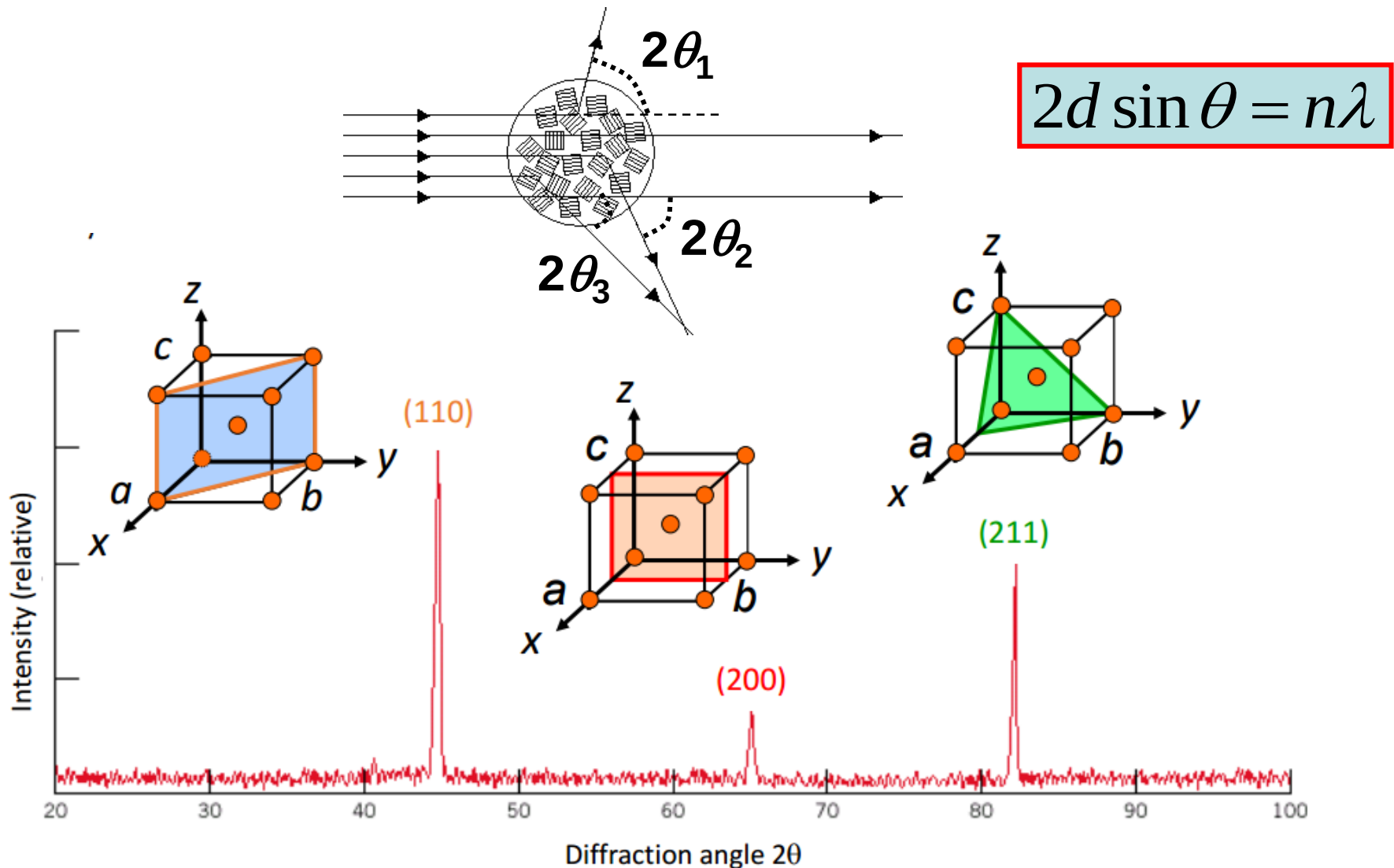
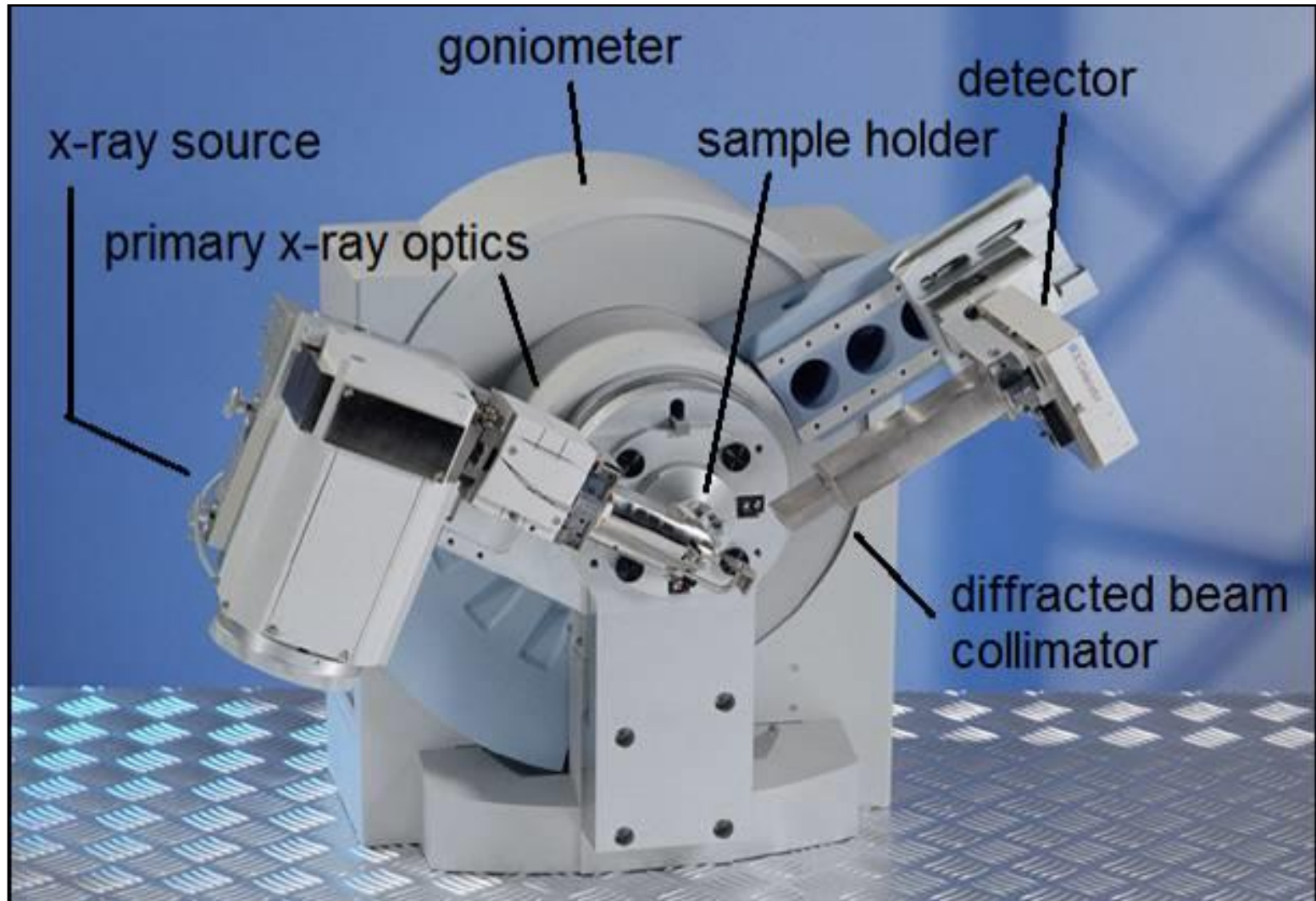


FIGURE 3.20 Diffraction pattern for polycrystalline α -iron.

XRD of Polycrystals



XRD of Polycrystals

For cubic systems
(SC, BCC, FCC)

$$d_{(hkl)} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$2d \sin \theta = n\lambda$$



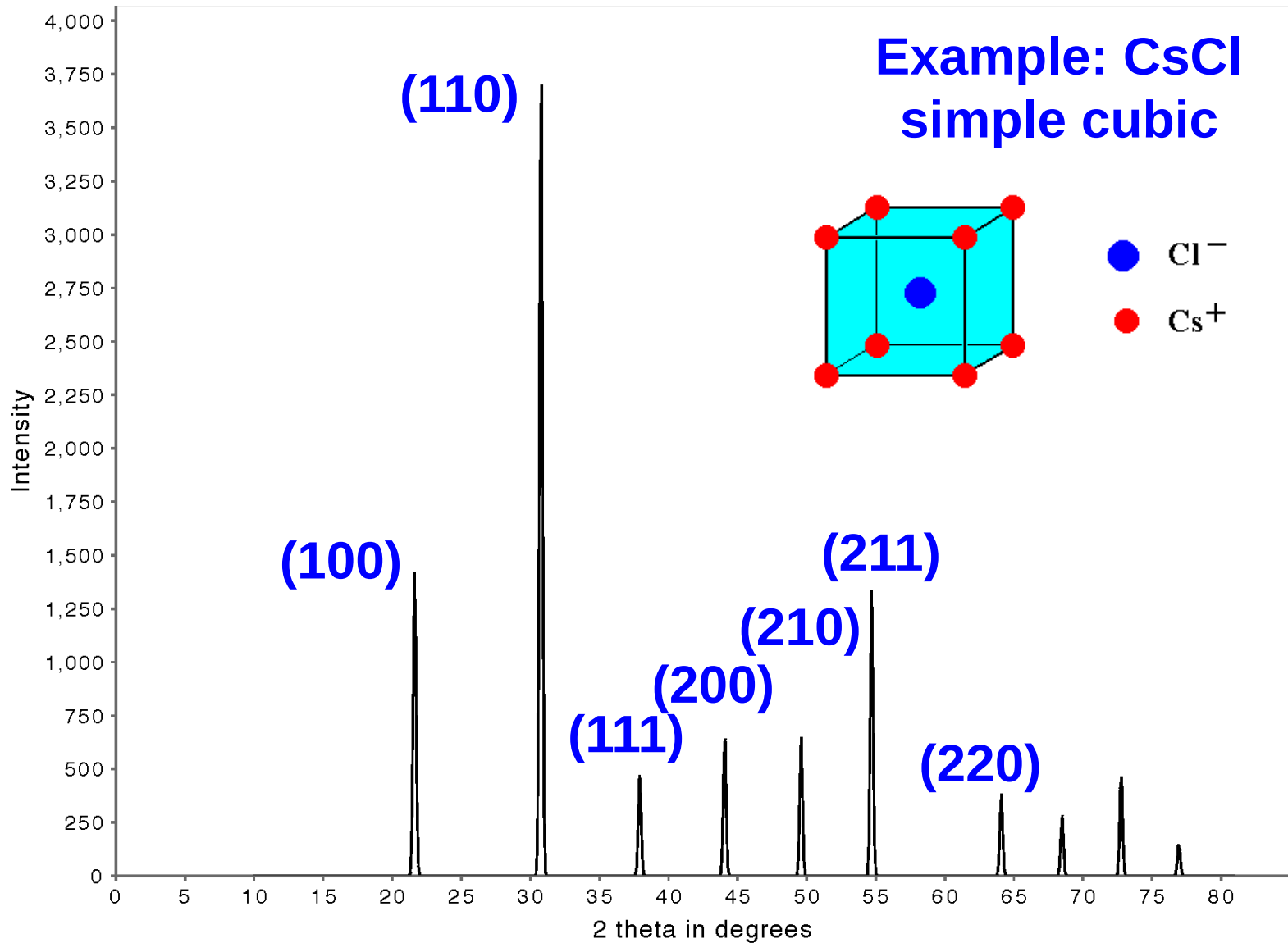
$$\sin^2 \theta = \frac{n^2 \lambda^2}{4a^2} (h^2 + k^2 + l^2)$$

angle	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	...
plane (hkl)	100	110	111	200	210	211	...

Typically, $n = 1$

Commonly used x-ray wavelength $\lambda = 0.154$ nm
generated by electrons hitting a copper metal

*-Cs Cl-[PM3-M]Cortona, P.[1992]



XRD of Polycrystals

- What determines the amplitude of the peaks?
- Why are some peaks missing?
 - Example: no (100) and (111) peaks for α -Fe

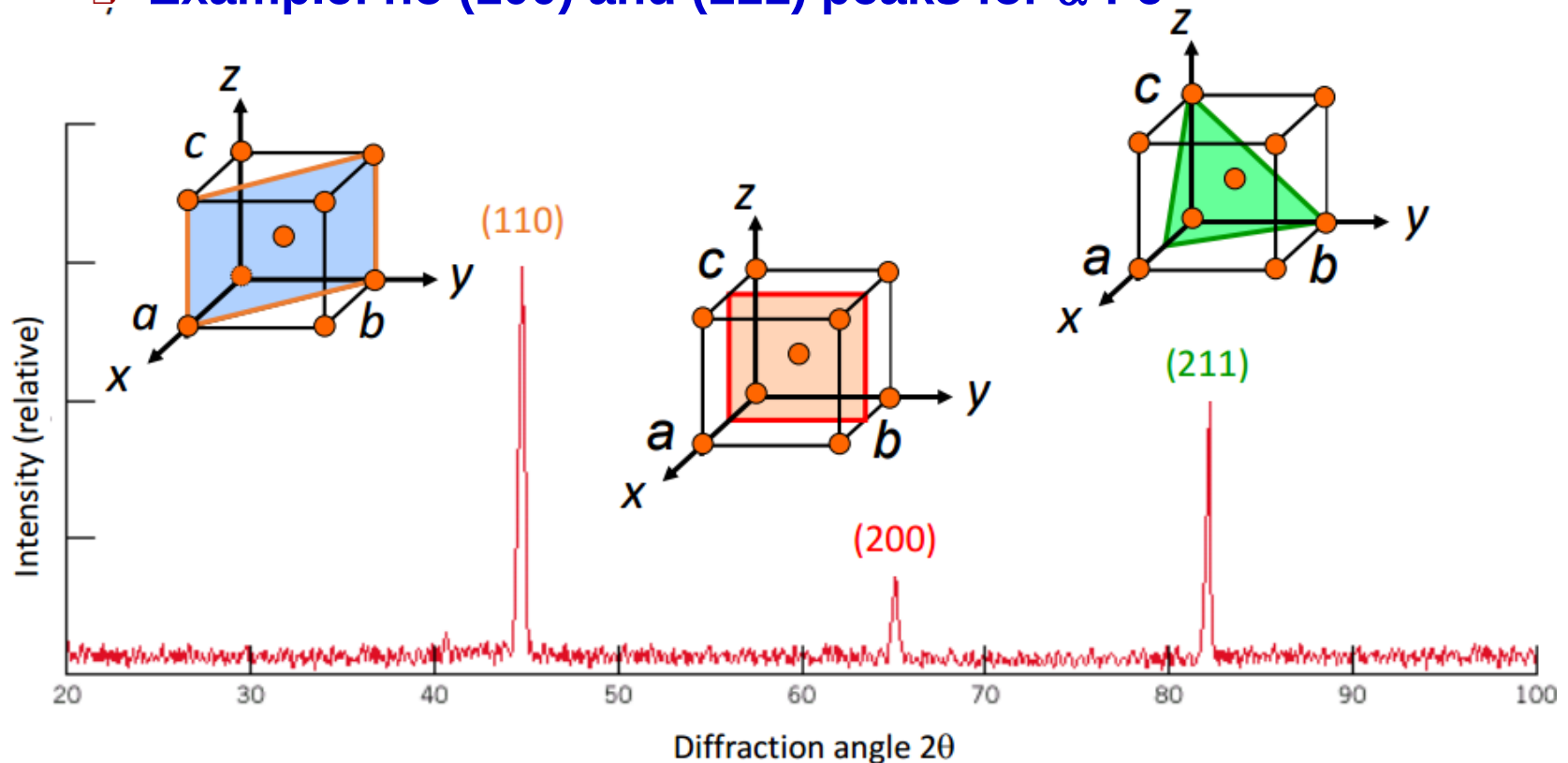


FIGURE 3.20 Diffraction pattern for polycrystalline α -iron.

XRD of Polycrystals

- What determines the amplitude of the peaks?
 - ❑ geometrical structure factor S (几何结构因子)
 - ❑ atomic form factor f (原子形状因子)
 - ❑ crystal orientation
 - ❑ crystal size
 - ❑ ...

Geometrical Structure Factor S

■ Systematic Absences 系统消光

- when $S = 0$, peaks are absent
- destructive interference

Ashcroft & Mermin
p.104–p.107

■ Allowed index

- SC

$h, k, l = \text{all integers}$

- BCC

$h + k + l = \text{even}$

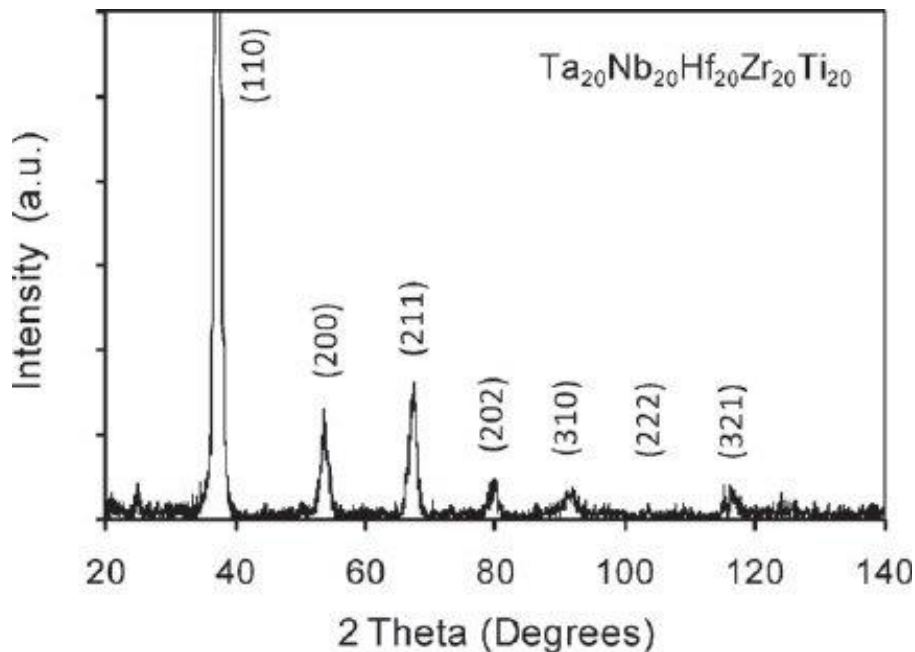
- FCC

h, k, l are all even or odd

Allowed list of $h^2 + k^2 + l^2$ for cubic crystals

Forbidden numbers	Primitive, P	Face Centered, F	Body Centered, I	Corresponding hkl
	1			100
	2		2	110
	3	3		111
	4	4	4	200
	5			210
	6		6	211
7				
	8	8	8	220
	9			221, 300
	10		10	310
	11	11		311
	12	12	12	222
	13			320
	14		14	321
15				
	16	16	16	400

Geometrical Structure Factor S

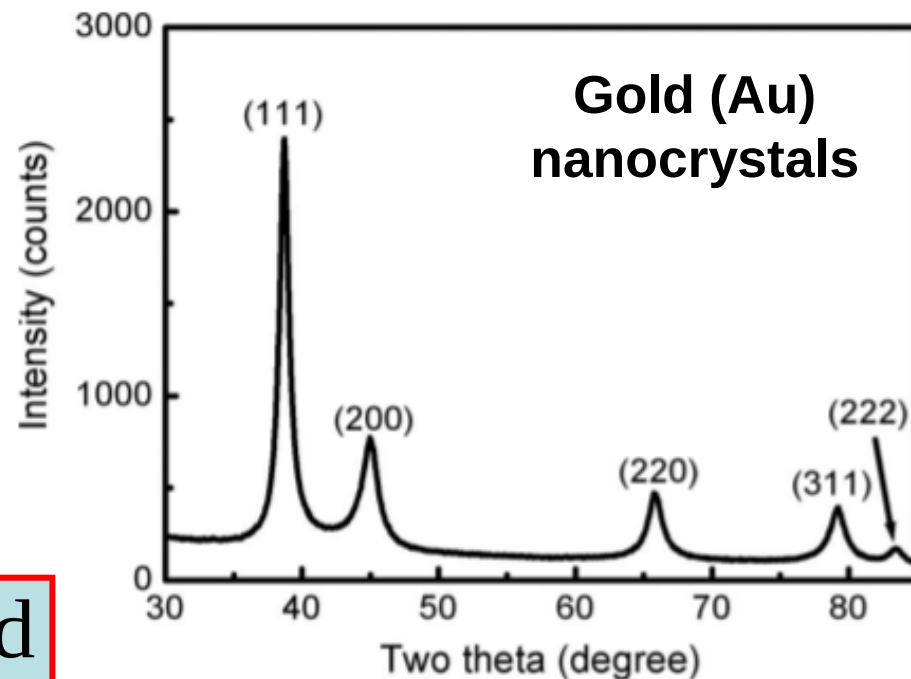


BCC

$$h + k + l = \text{even}$$

FCC

$$h, k, l \text{ are all even or odd}$$



Geometrical Structure Factor S

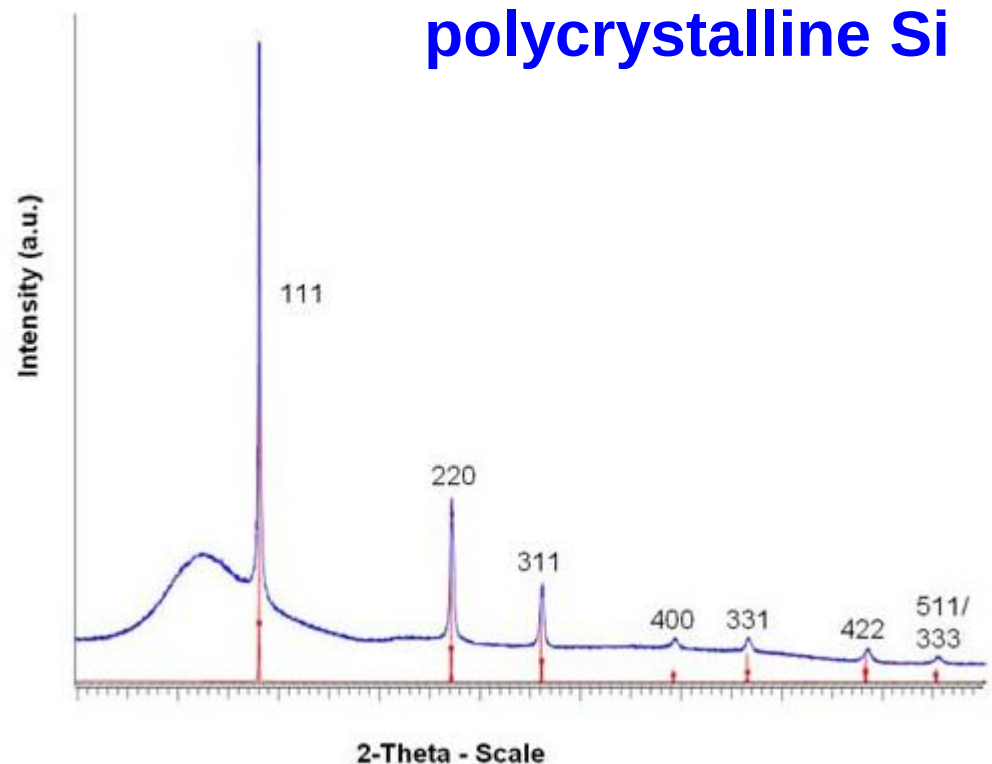
- Allowed index for **Diamond Structure (C, Si, Ge)**
 - it is FCC, but more special

h, k, l are all odd

$(111), (311), (331), \dots$

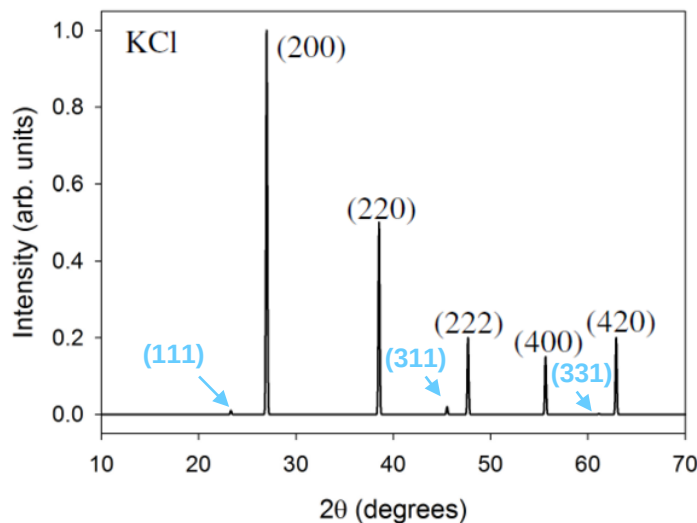
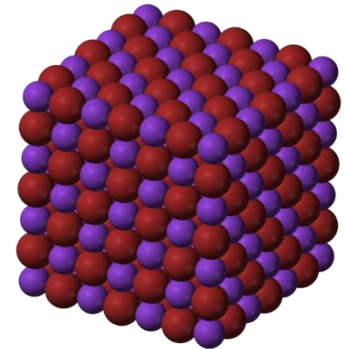
h, k, l are all even
and $h + k + l = 4n$

$(220), (400), (422), \dots$

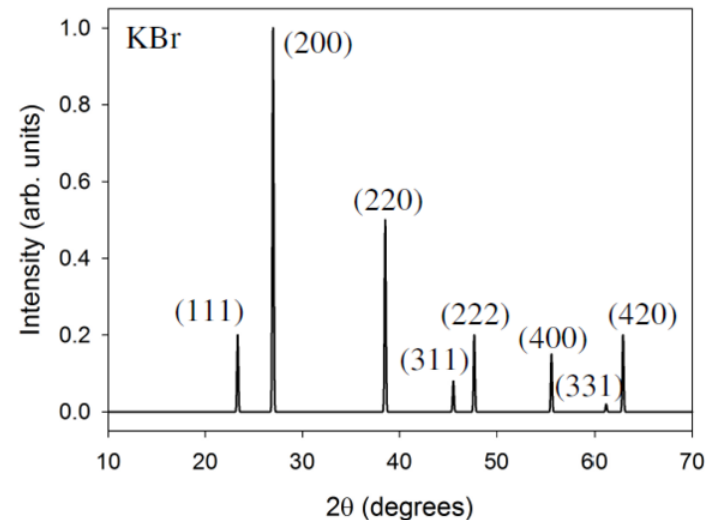


Atomic Form Factor f

- Example: KCl and KBr
 - FCC lattice, similar lattice parameter a



K^+ and Cl^- are very similar
KCl looks like a simple cubic with $a/2$

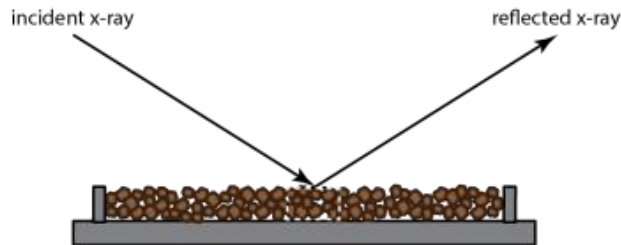


K^+ and Br^- are very different
KBr is a normal FCC with a

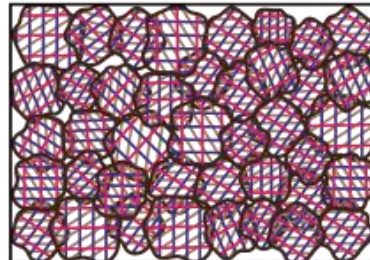
Crystal Orientation

- Aligned crystals show strong orientation preference

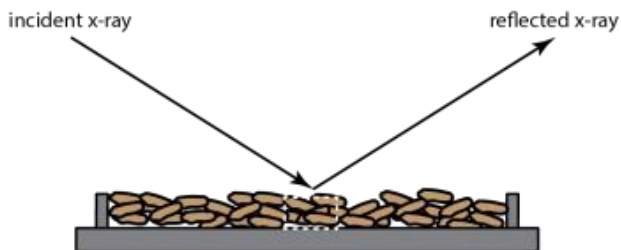
No Preferred Orientation



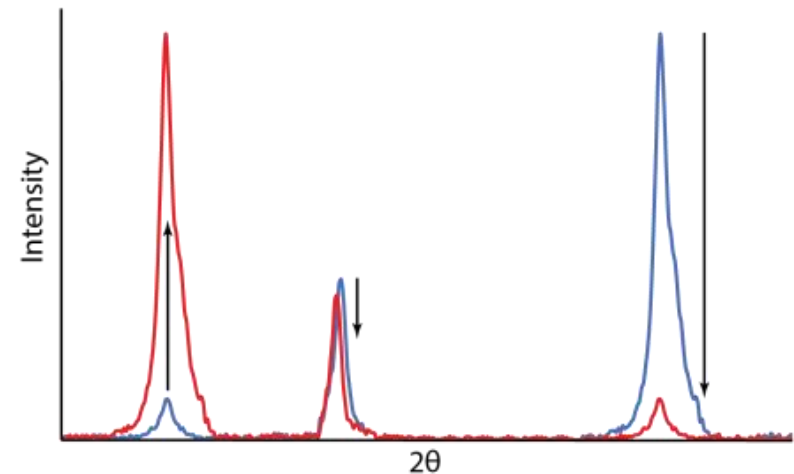
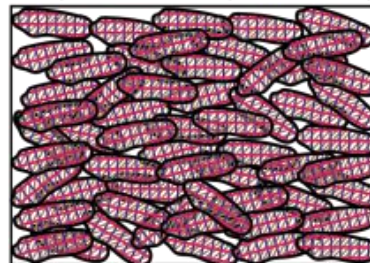
All planes detected equally



Preferred Orientation



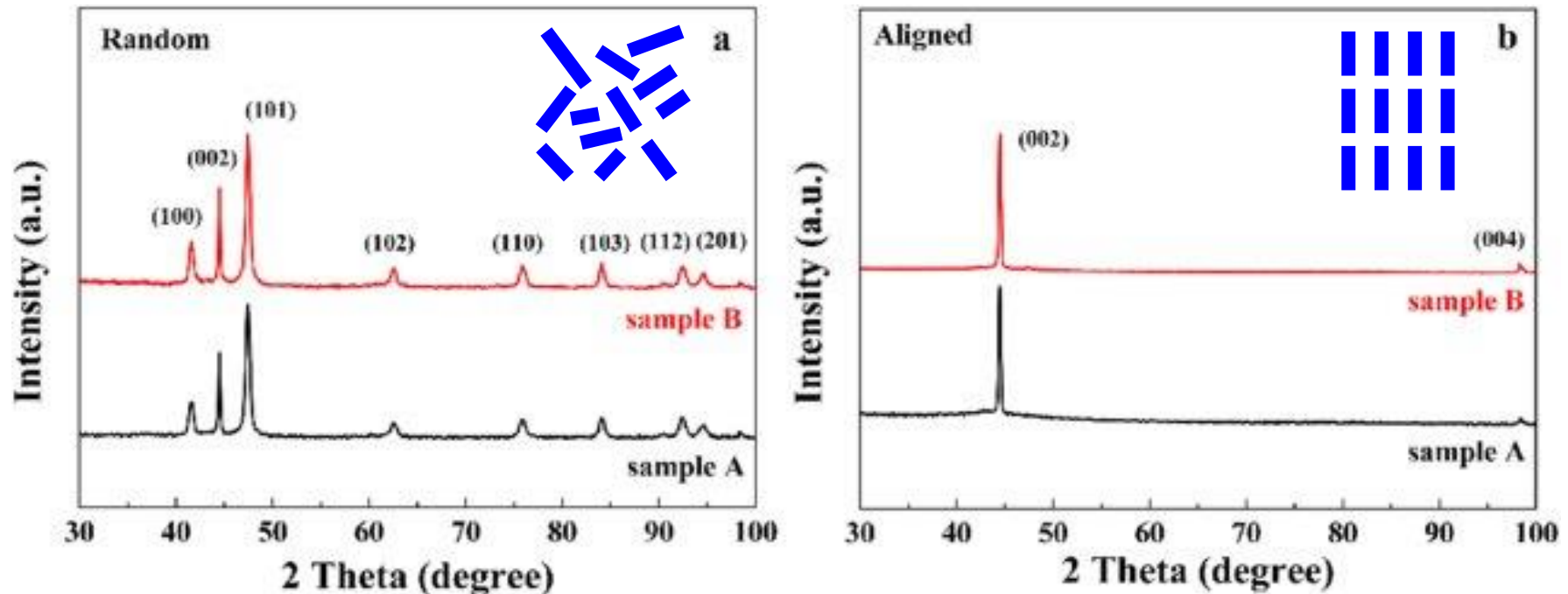
One plane detected preferentially



No preferred orientation - correct intensity ratios
Preferred orientation - highly skewed intensity ratios

Crystal Orientation

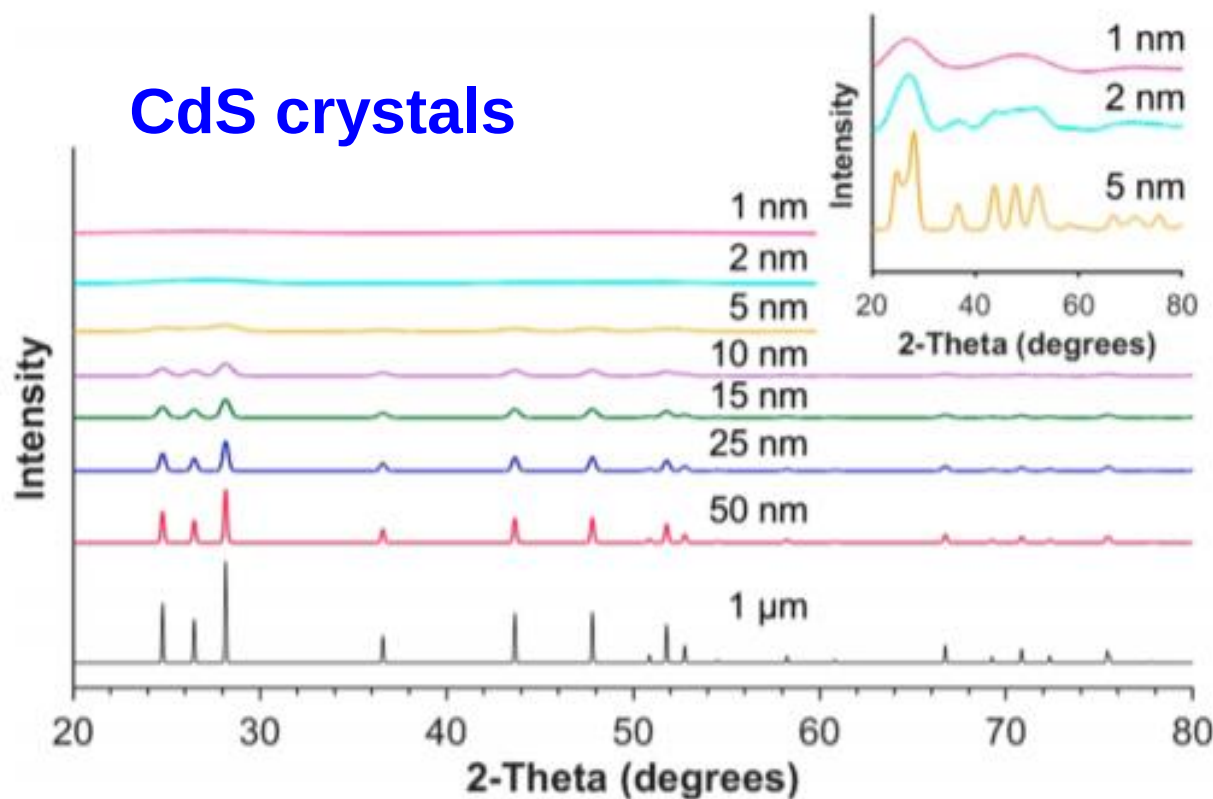
- Aligned crystals show strong orientation preference



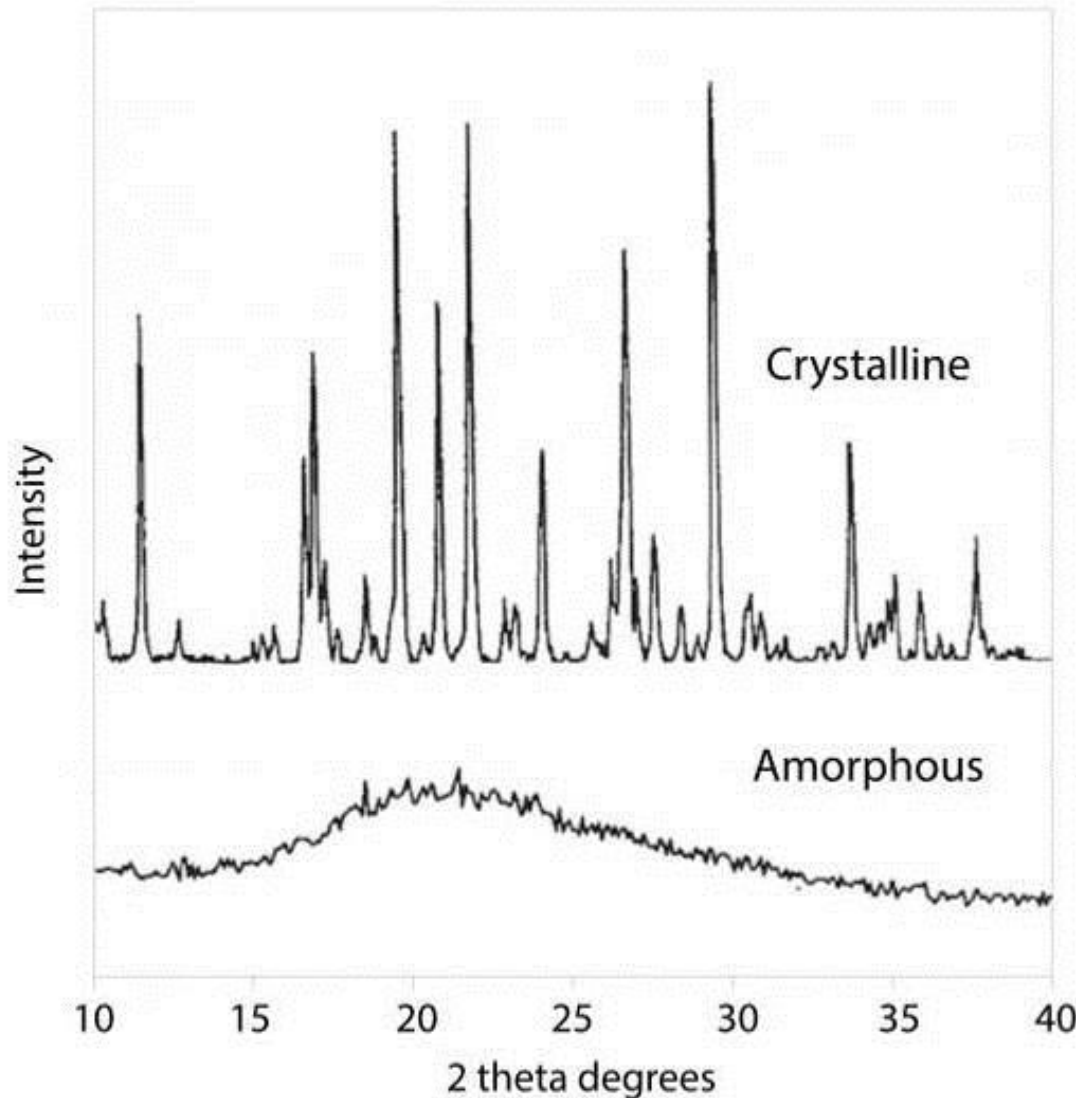
Co nanowires

Crystal Size

- Smaller crystals lead to lower diffraction intensity and larger peak width.



Amorphous Materials 非晶



Crystalline:
anisotropic (各向异性)

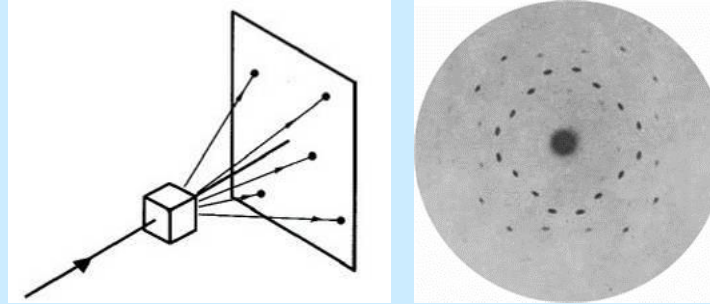


Amorphous:
isotropic (各向同性)



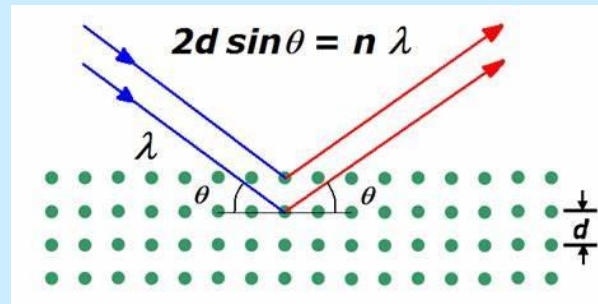
History of Discovery

**X-ray
diffraction
of crystals**



**M. von Laue (劳厄)
Nobel Prize in 1914**

Bragg's law



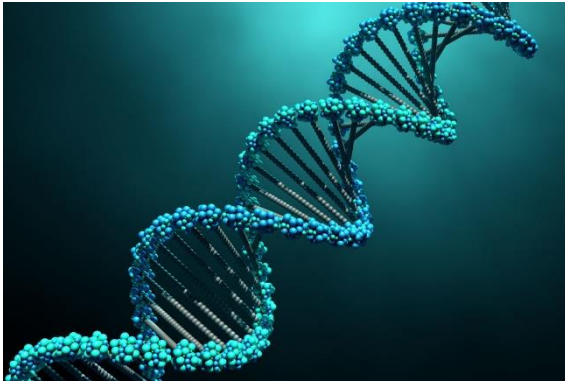
**Bragg & Bragg
(布拉格父子)
Nobel Prize in 1915**

Crystal Structure

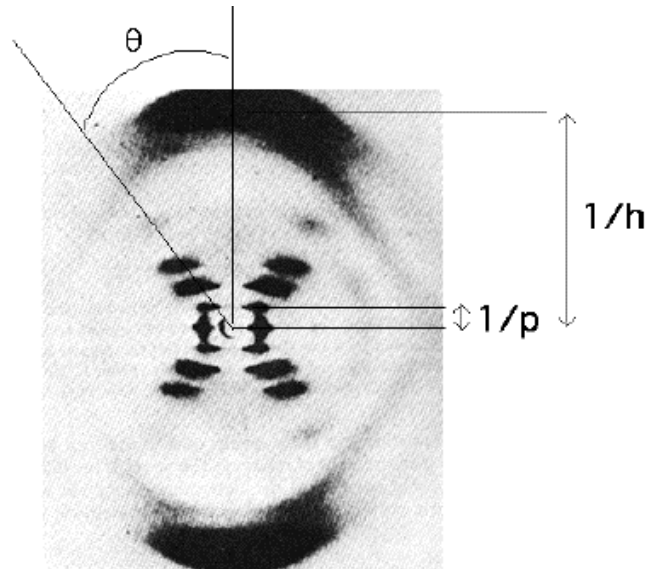


Diffraction Pattern

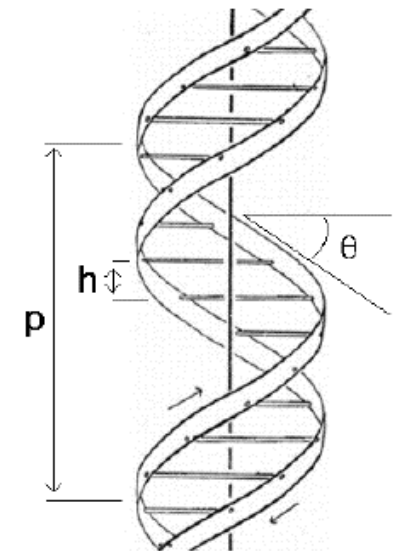
The Structure of DNA



XRD pattern



DNA structure



$$p = 3.4 \text{ nm}$$

$$h = 0.34 \text{ nm}$$

$$\theta = 32^\circ$$



J. Waston, F. Crick and M. Wilkins

**1962 Nobel Prize in
Physiology / Medicine**



**R. Franklin
died in 1958**

J. Watson, F. Crick, *Nature* **171**, 737 (1953)
M. Wilkins, et al., *Nature* **171**, 738 (1953)
R. Franklin, R. Gosling, *Nature* **171**, 740 (1953)

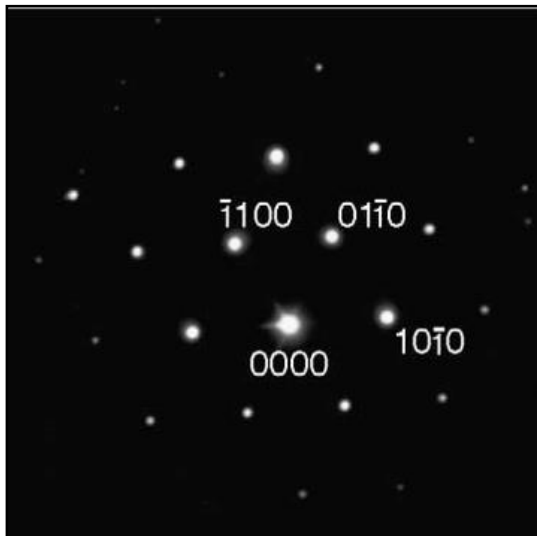
Electron Beam Diffraction

- Electron Beam is also a wave
 - ▣ also follows Bragg's law

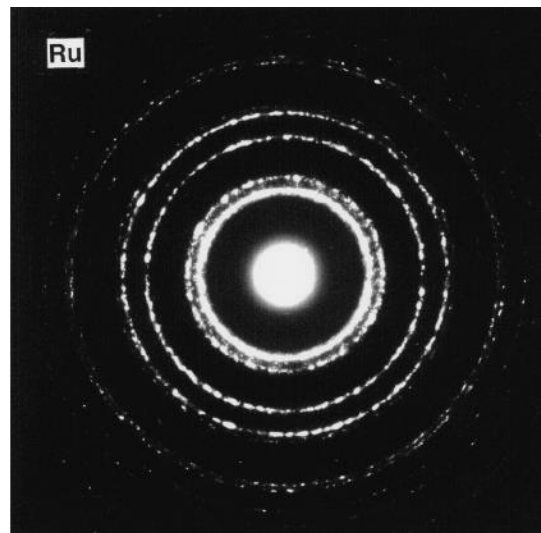
wave-particle duality

$$\lambda = \frac{h}{\text{momentum}}$$

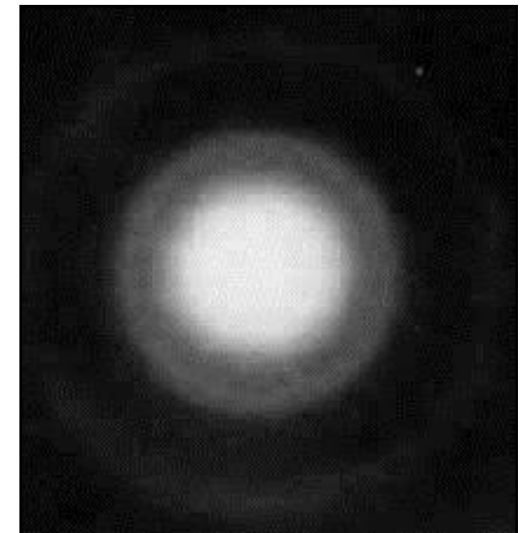
diffraction patterns



single crystal



polycrystal

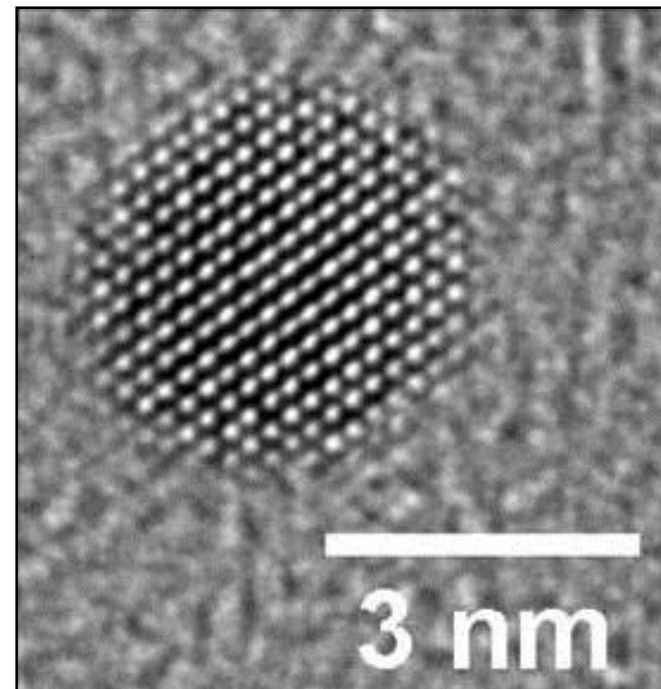
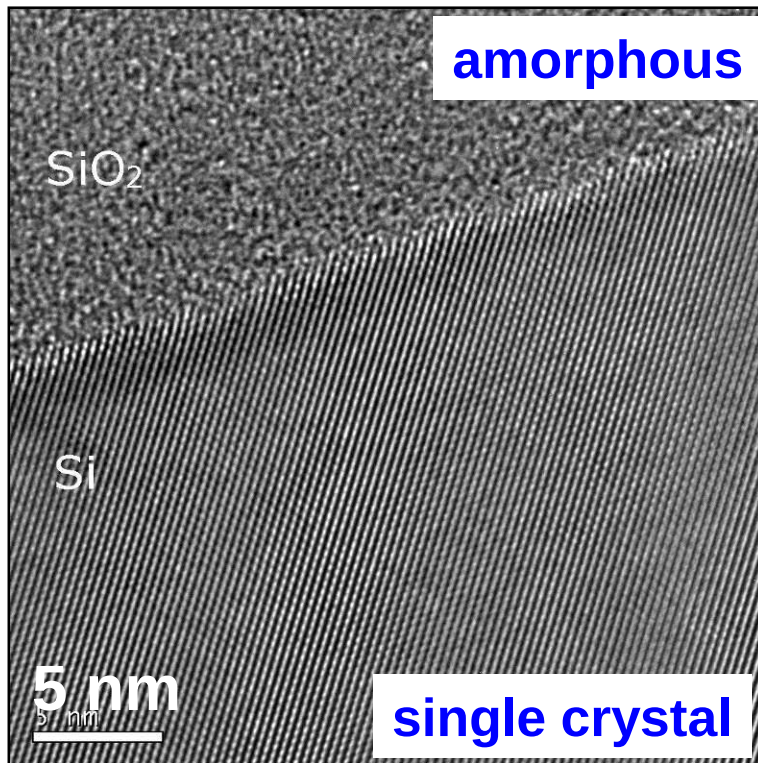


amorphous

Electron Microscopy

■ HRTEM

▣ High Resolution Transmission Electron Microscope



quantum dot

Neutron Diffraction 中子衍射

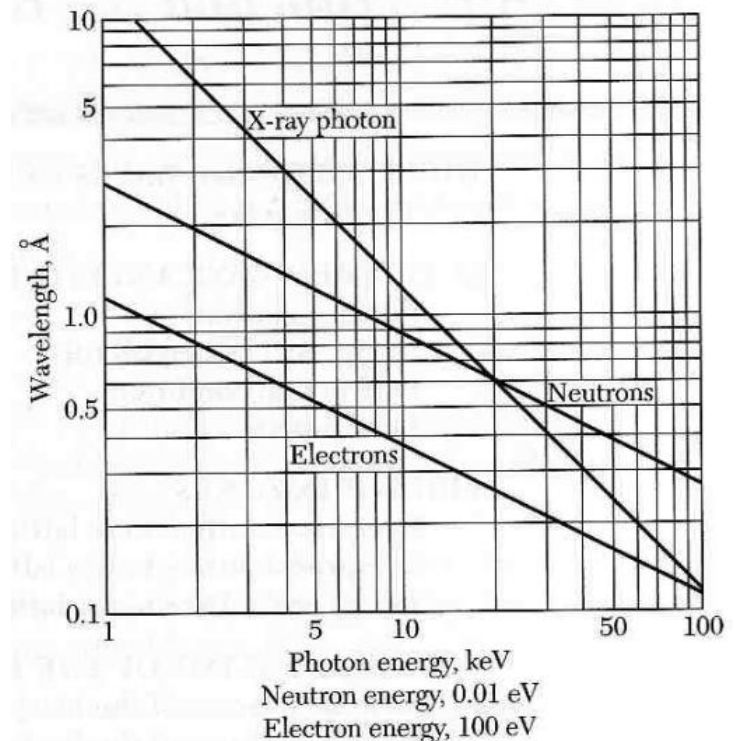
- Neutron Beam is also a wave
 - ▣ Lower energy
 - ▣ No charge
- Need a nuclear reactor

wave-particle duality

$$\lambda = \frac{h}{\text{momentum}}$$



散裂中子源，东莞



This Class

- Introduction (Week 1)
- **Materials and Crystal Structures (Week 2–3)**
 - ▣ Crystal structures, lattices
 - ▣ Reciprocal space, Brillouin zones
 - ▣ Materials Characterization: Wave diffraction, the Bragg law
- Electronic Properties (Week 4–12)
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

Next Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

Thank you for your attention